

A Brief Derivation of Spacetime

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The problem

Why does time have a direction that space does not?

The arrow of time is one of the fundamental open problems in physics. Thermodynamics traces it to initial conditions (Boltzmann, 1872). Cosmology to a boundary condition (Penrose, 1979). Quantum mechanics to measurement (von Neumann, 1932). Each adds structure beyond the equations to break the symmetry.

The arrow was a sign. Space is positive: every distance a sum of squares. Time is positive too: $t > 0$, $c > 0$, $\Lambda > 0$. Both are observed. This paper shows they are one algebra in two signatures.

Six steps from dot product to spacetime

Step 1. The metric.

Space carries a dot product $\langle x, y \rangle = x_1y_1 + x_2y_2 + \cdots + x_dy_d$, where d is the dimension. The components are the Euclidean metric δ_{ij} . All distances are positive, no direction is special.

Step 2. Translations.

The symmetries of δ_{ij} are rotations J_{ab} and translations P_a . On flat space, translations commute: shifting east then north is the same as north then east.

On a curved surface they do not: walking east then north on a sphere does not return you to the same point as north then east. The failure to commute is curvature. Writing $[P_a, P_b] = P_aP_b - P_bP_a$ for the difference, the three maximally symmetric geometries, those preserving the full rotational symmetry of δ_{ij} , are classified by one number:

$$[P_a, P_b] = \lambda J_{ab}.$$

λ is the curvature. $\lambda > 0$: sphere. $\lambda = 0$: flat. $\lambda < 0$: hyperbolic.

Step 3. Finiteness.

The metric δ_{ij} determines a volume form on each of the three geometries. The sphere S^d is compact and has finite total volume; $\mathbb{R}^d$ and H^d are non-compact with infinite total volume. A normalized measure invariant under the full isometry group requires external structure on $\mathbb{R}^d$ and H^d (a cutoff, boundary condition, or weight function) that δ_{ij} does not contain. $\lambda > 0$ is forced.

Step 4. Time.

The sphere S^d sits in $\mathbb{R}^{d+1}$ as $x_1^2 + \cdots + x_{d+1}^2 = R^2$, with isometry algebra $\mathfrak{so}(d+1)$. Wick rotating one coordinate, $x_{d+1} \rightarrow ix_0$, takes the sphere to the hyperboloid $x_1^2 + \cdots + x_d^2 - x_0^2 = R^2$ and the algebra to $\mathfrak{so}(d, 1)$. Two signatures for the extra direction, two real algebras with the same spatial $\mathfrak{so}(d)$ and the same λ . The hyperboloid is de Sitter space, with cosmological constant $\Lambda = \frac{1}{2}(d-1)(d-2)\lambda$ (in $d = 4$,

$\Lambda = 3\lambda$). The algebra $\mathfrak{so}(d, 1)$ is simple and rigid: by Whitehead, no smooth deformation. The sign of λ is fixed by the choice of signature; the magnitude is a scale.

Step 5. The matrix.

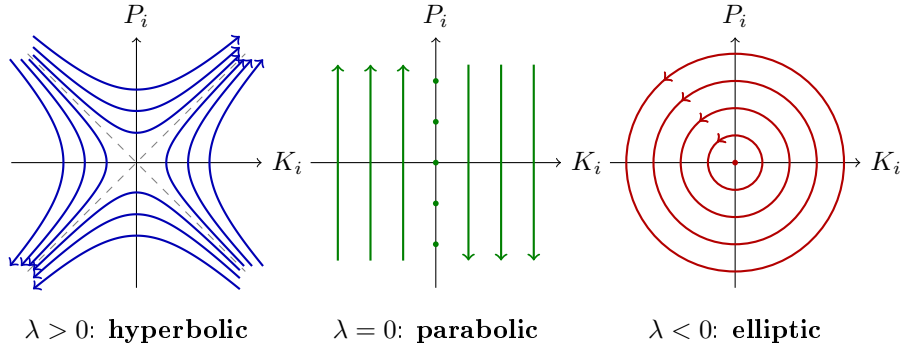
Realize $\mathfrak{so}(d, 1)$ as the antisymmetric matrices preserving $\eta = \text{diag}(+1, \dots, +1, -1)$ on $\mathbb{R}^{d+1}$. Identify rotations $J_{ij} = M_{ij}$, boosts $K_i = M_{di}$, time translation $P_0 = M_{0d}/L$, spatial translations $P_i = M_{0i}/L$, where $L = 1/\sqrt{\lambda}$. The standard $\mathfrak{so}(d, 1)$ commutators give boosts $[K_i, P_j] = \delta_{ij}P_0$ and $[K_i, P_0] = P_i$: an invariant scale, the speed of light c , and the Lorentz subalgebra of special relativity. Time evolution is the adjoint action of P_0 : $[P_0, K_i] = -P_i$ and $[P_0, P_i] = -\lambda K_i$. On each (K_i, P_i) plane, this is a 2×2 matrix:

$$\text{ad}(P_0) = \begin{pmatrix} 0 & -\lambda \\ -1 & 0 \end{pmatrix}.$$

For $\lambda > 0$ the eigenvalues $\pm\sqrt{\lambda}$ are real and distinct, with eigenvectors $\ell_{\pm} = P_i \mp \sqrt{\lambda}K_i$ at eigenvalue $\pm\sqrt{\lambda}$. The flow is hyperbolic:

$$e^{t \text{ad}(P_0)} = \cosh(\sqrt{\lambda}t) I + \frac{\sinh(\sqrt{\lambda}t)}{\sqrt{\lambda}} \text{ad}(P_0).$$

In pure de Sitter, the rate $\sqrt{\lambda}$ is the Hubble constant $H = \sqrt{\Lambda/3}$ (in $d = 4$), the observed rate of cosmological expansion.



*How time evolution acts in the (K_i, P_i) plane for each sign of λ .
Only $\lambda > 0$ separates. The other two are shown for comparison.*

Step 6. The sign.

Time evolution is a symmetry of the geometry, so it preserves spacetime length. It scales one eigendirection by $e^{+\sqrt{\lambda}t}$ and the other by $e^{-\sqrt{\lambda}t}$. A length-preserving flow can scale a vector by $s \neq 1$ only if its length is zero; otherwise the length would change by s^2 . In spacetime (distance squared minus time squared) some non-zero vectors have zero length: light rays. The eigendirections are light rays. The exponential scaling is the cosmological redshift; the scale factor $e^{\sqrt{\lambda}t}$ satisfies the Friedmann equation $H^2 = \lambda$.

One eigendirection grows without bound; this is the future horizon, the cosmological horizon beyond which signals never reach us. The other contracts toward zero; this is the past horizon, the limit beyond which light has not yet arrived. They are different generators in the algebra.

The eigenvalues are signed: $+\sqrt{\lambda}$ and $-\sqrt{\lambda}$. The positive eigenvalue is $t > 0$. It is the sign of cosmological expansion, the sign of c , the sign of Λ . The same sign.

The future is the direction of positive t increase, just as positive Λ means the universe expands in time. $t > 0$. □